Topic 12 – Governed, Optimized Workflow

Implementation Plan

Project Name: Governed Enterprise Intelligence Assistant
Version: 1.0
Implementation Platform: Google Opal

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1. Implementation Overview

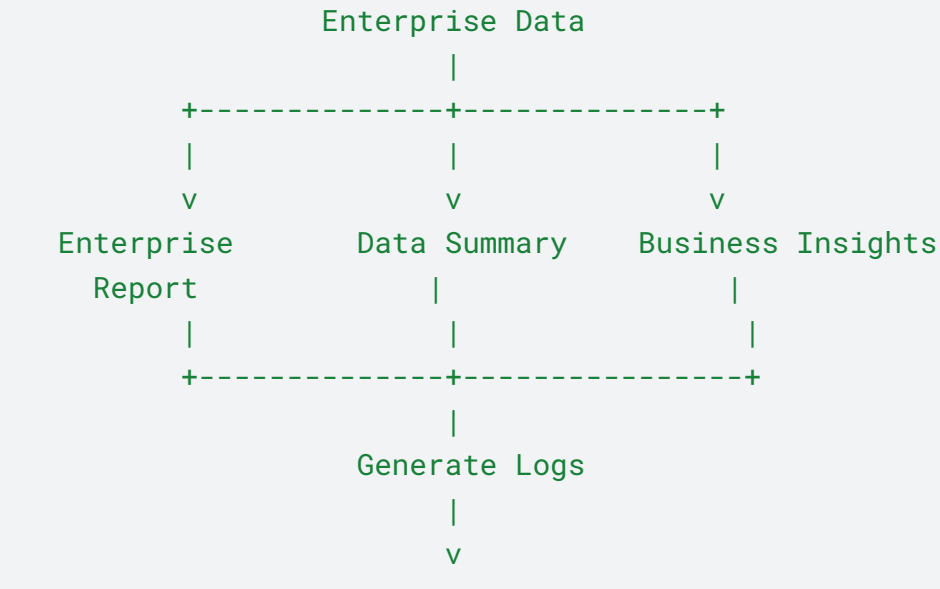
The project is implemented as a multi-step AI workflow in Google Opal. It accepts enterprise data, processes the input through independent generation tasks, compiles the generated information into an executive portal, and validates a manager's approval decision.

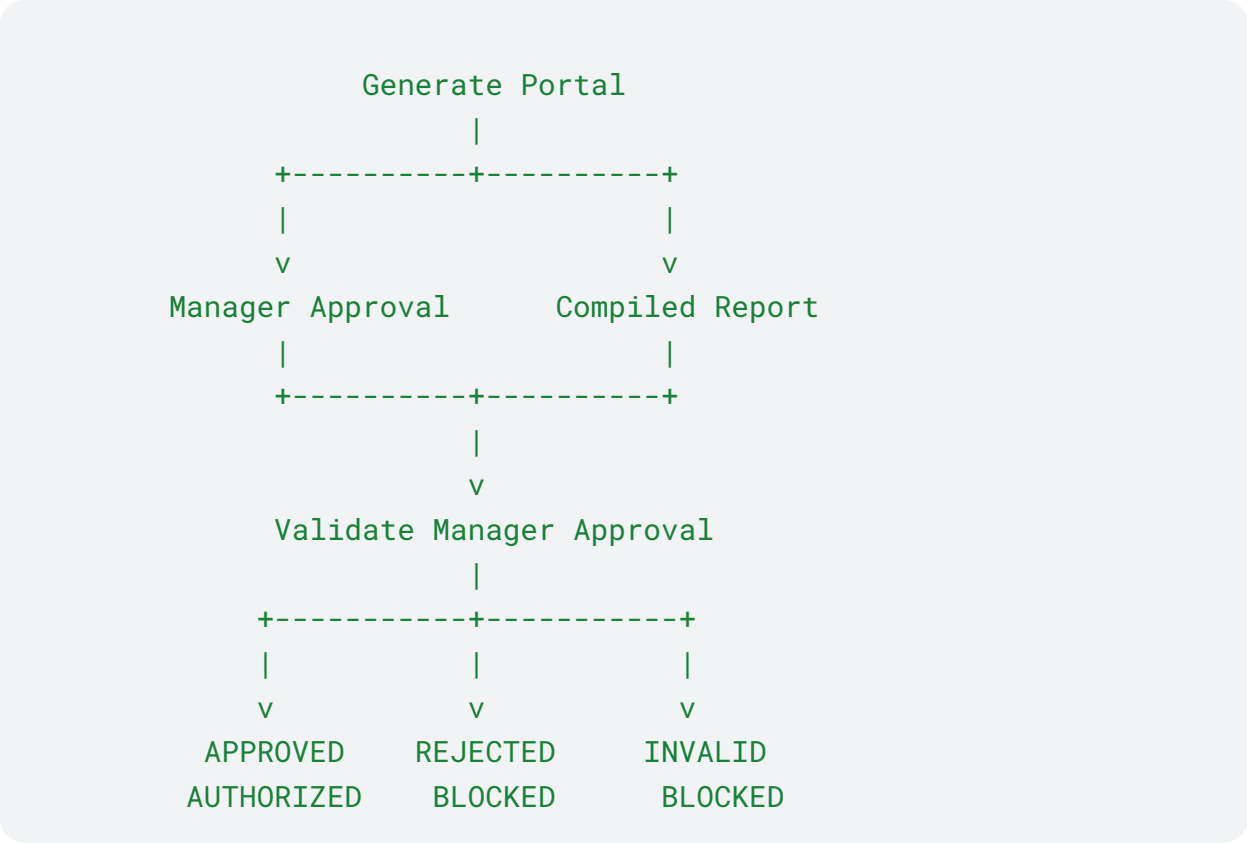
The implementation prioritizes parallel processing, monitoring, and human-in-the-loop governance.

2. Workflow Architecture

The workflow follows this logical structure:

None





Important: This diagram represents the logical design. The current Opal workflow implements the connected generation and validation steps, but does not implement automatic conditional routing or Google Drive saving.

3. Opal Workflow Components

Component	Opal Step	Responsibility
Input	Enterprise Data	Receives the enterprise source document.
Monitoring	Generate Logs	Prepares workflow execution and audit information.
Generation	Generate Enterprise Report	Creates the structured enterprise report.
Summarization	Generate Data Summary	Extracts key metrics and business trends.
Analysis	Extract Business Insights	Identifies insights, risks, and opportunities.

Compilation	Generate Portal	Compiles the available outputs into an executive review portal.
Human Input	Manager Approval	Collects the manager's decision.
Validation	Validate Manager Approval	Validates the decision and determines authorization status.

4. Implementation Steps

Step 1 – Create the Opal Workflow

Create a new Opal application named:

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Governed Enterprise Intelligence Assistant
```

Add an Enterprise Data input step to accept the source document.

Step 2 – Configure Parallel Generation

Create the following independent generation steps:

1. Generate Enterprise Report.
2. Generate Data Summary.
3. Extract Business Insights.

Connect Enterprise Data directly to each generation step so that they can independently process the same source.

Step 3 – Configure Monitoring

Create the Generate Logs step.

Its prompt should prepare a structured monitoring record containing:

- Workflow name.
- Execution start timestamp.
- Enterprise input received.
- Individual generation task statuses.
- Completion timestamp.
- Available execution latency.
- Errors and warnings.
- Overall workflow status.

Use available memory functionality where configured. Persistent centralized storage is not confirmed.

Step 4 – Configure Executive Portal

Create the Generate Portal step.

Connect these four upstream outputs:

- Generate Enterprise Report.
- Generate Data Summary.
- Extract Business Insights.
- Generate Logs.

Configure the portal to display the executive summary, business performance, insights, monitoring information, and pending approval status.

Step 5 – Configure Human Approval

Create the Manager Approval input step.

Ask the manager to submit one explicit decision:

- APPROVE
- REJECT

The prompt should instruct the manager not to assume approval and should treat missing or unclear input as invalid.

Step 6 – Configure Approval Validation

Create the Validate Manager Approval step.

Connect:

- Manager Approval.
- Generate Portal.

Configure validation rules:

Input	Output	Authorization
APPROVE	APPROVED	AUTHORIZED
REJECT	REJECTED	BLOCKED

Invalid input	INVALID	BLOCKED
Blank input	INVALID	BLOCKED

The validation agent must not save or modify the report.

Step 7 – Configure Sharing Restrictions

Open the Opal sharing settings.

Use the available restricted sharing option rather than public link sharing. Grant access only to intended reviewers, using the appropriate viewer permissions where available.

Step 8 – Preview and Test

Use Preview to verify:

- Enterprise input processing.
- Generation outputs.
- Executive portal compilation.
- Data accuracy.
- Monitoring output.
- Manager approval validation.

Record the actual outputs and errors in the Testing Results Document.

5. Governance Implementation – 5:3:1

Monitoring – 5

The Generate Logs step prepares monitoring information for workflow execution and generation task statuses.

Actuation – 3

The approval validation step determines whether a controlled saving action may be authorized. A real save operation must be implemented separately and must not be assumed to exist.

Human-in-the-Loop – 1

The Manager Approval step collects an explicit decision. The validation step blocks rejected, invalid, and blank decisions.

6. Testing Performed

The following approval scenarios were tested:

- TC-01: APPROVE – PASS.
- TC-02: REJECT – PASS.
- TC-03: MAYBE – PASS.
- TC-04: Blank approval – PASS.

Additional checks covered workflow connections, portal sections, source figures, monitoring output, and pending approval status.

7. Implementation Limitations

The following items remain pending:

1. Google Drive saving: no suitable save action was available in the current Opal editor.
2. Conditional routing: no conditional branch tool was available in the current editor.
3. Persistent audit storage: central storage of logs has not been confirmed.
4. Enterprise RBAC: full role-based access enforcement has not been independently verified.
5. Opal Console debugging: a complete console-based debugging process has not been verified.

8. Future Enhancements

- Integrate an approved Google Drive document-creation or file-saving action.
- Implement approval-based conditional routing when supported.
- Store audit records in an authorized persistent data store.
- Enforce role-based permissions for reviewers and administrators.
- Add execution monitoring and failure alerts.
- Perform integration and end-to-end tests after the missing capabilities are configured.

9. Final Implementation Status

Implemented: Core input, parallel generation connections, report compilation, monitoring output, manager approval, and approval validation.

Pending: Automated Google Drive saving, conditional routing, persistent centralized audit storage, and verified enterprise RBAC.

Overall Status: Core Workflow Implemented – Partial Implementation.

10. Conclusion

The Governed Enterprise Intelligence Assistant demonstrates the main workflow-generation and approval-governance concepts using Google Opal. The implementation provides a foundation for a governed enterprise reporting workflow, with further integrations required before it can be considered a fully automated enterprise solution.